

OMAR AL ETHAMAT

Computer Vision and Robotics Engineer

Eligible to work in the UK | omaraethamat@gmail.com | [Portfolio](#) | [Github](#) | [Linkedin](#)

AI & Robotics Engineer focused on Computer Vision, Robotics, and autonomous intelligent systems, with hands-on experience developing real-world AI and embedded applications. Passionate about building practical technologies that combine software, hardware, sensors, and robotics into functional systems, including space and advanced engineering applications. Experienced in designing end-to-end solutions through experimentation, system integration, and applied AI engineering.

PROFESSIONAL EXPERIENCE

AI Engineer Intern, Orange Jordan . Internship

Feb 2026 - Present

- Applied Computer Vision and image preprocessing techniques for real-world AI applications
- Developed an end-to-end Speech-to-Text (STT) service, including model development, backend APIs, and production deployment
- Built and optimized scalable backend and data systems, including RAG pipelines, database integration, and large-scale data processing using Apache Spark

Software Engineer, Al-Rhim Sat . Part-time

Nov 2025 - Present

- Developing embedded software for a student-led 1U CubeSat selected for launch through United Nations Office for Outer Space Affairs (UNOOSA) and Exolaunch initiative
- Programming the On-Board Computer (OBC), handling data processing and satellite communication functions
- Designing and implementing microcontroller-based systems for wildlife tracking using LoRa technology in real-world environments

Computer Vision Engineer , Qynerva . Full-time

April 2025 - Present

- Developed multi-stage deep learning pipeline (segmentation, classification, XAI)
- Achieved 96.77% accuracy (F1: 0.9679) across 4 tumor classes (100% on normal)
- Built segmentation model with 98.98% voxel accuracy (Dice up to 0.78)
- Designed explainable AI using visual heatmaps
- Developed end-to-end platform with backend + chatbot
- Applied generative AI techniques to synthesize MRI data

Machine Learning Virtual Intern , Intern Intelligence . Internship

Feb 2025 - March 2025

- Developed 3+ ML/NLP models (cybersecurity, healthcare, text classification) achieving up to 98% accuracy in spam detection and 96%+ in medical classification
- Built end-to-end pipelines including preprocessing, TF-IDF/LSTM modeling, and hyperparameter tuning (GridSearchCV), improving model performance by 10–15%
- Deployed interactive apps using Streamlit for real-time prediction and moderation, handling multi-class classification with 87% accuracy

NLP Engineer , High Yield Medicine . Freelance

Feb 2025 - March 2025

- Developed an NLP-based medical research retrieval system, achieving 100% recall (no missed relevant studies) and 79% precision when benchmarked against expert annotations
- Integrated LLMs and external APIs to improve search relevance, reaching an F1-score of 0.88 and increasing the accuracy of literature discovery

EDUCATION

B.Sc. in Artificial Intelligence and Robotics

Oct 2022 - Oct 2026

Al-Balqa Applied University (BAU)

Software Engineering Program (Core Curriculum)

Aug 2025 - Oct 2026

42 Amman

- Project-based training in C/C++, system programming, networking, and software engineering

CERTIFICATIONS

Sequence Models DeepLearning.AI	May 2025
Neural Networks and Deep Learning DeepLearning.AI	May 2025
Back-end Programming for Web Applications Al-Balqa Applied University (BAU)	Apr 2025
Backend Development Course IEEE BAU CS (Computer Society)	Oct 2024
Azure Fundamentals Microsoft	Mar 2025
Certificate of Participation - 1st At University Rank IEEEXtreme	Oct 2024

SKILLS

AI: ML, DL, Computer Vision, Generative AI, Reinforcement Learning, OpenCV, TensorFlow, PyTorch, YOLO, CNNs
Programming: Python, C/C++,
Data & Analysis: Data Analysis, Databases, Pandas, NumPy, Apache Spark
Tools & Technologies: Linux, Git, Docker, Microsoft Azure
Embedded & Robotics: Embedded Systems, Microcontrollers, ROS, Arduino
Networking & Systems: Networking, System Administration, DevOps

PROJECTS

EduVision — Smart School System — Graduation Project Link - Developed a multi-modal AI system for real-time student monitoring using three integrated modules: Camera, Smartwatch, and Academic Performance analytics - Built a YOLO-based Computer Vision model for student engagement and classroom activity analysis, achieving 77% mAP - Developed a MediaPipe Face Mesh pipeline to analyze focus and distraction through facial movement and head-position tracking - Built smartwatch and academic prediction modules for physiological and performance analysis, achieving up to 91% and 89% accuracy respectively - Implemented live dashboards, historical analytics, and AI-generated reports with chatbot integration for real-time student monitoring - Performed end-to-end system validation using custom ground-truth testing, achieving 100% accuracy in controlled scenarios
VisionPong — Computer Vision Game Link - Built a real-time hand gesture-controlled game using computer vision techniques - Implemented object tracking and interaction using OpenCV and MediaPipe
MiniShell — System Programming Project Link - Built a Unix-like shell in C, implementing core Bash functionalities including pipes, redirections, and environment variable expansion - Designed a custom lexer and parser to handle command tokenization, quoting, and syntax interpretation - Managed process execution using fork/execve, file descriptors, and signal handling (Ctrl-C/D), ensuring stable multi-process behavior
OrbitCV — Satellite Image Processing Tool Link - Built an interactive satellite image preprocessing tool using C++ and OpenCV for real-time image transformations and analysis. - Implemented image operations including cropping, rotation, scaling, affine, and perspective transformations with keyboard and mouse interaction. - Integrated a practical SDR-based satellite reception workflow, processing real satellite images captured through custom antenna setups.